Credit Risk Portfolio Analytics Platform

Complete Technical, Methodological & Empirical Guide: Architecture, Key Concepts, Default Modeling, Insights & Dual AI Governance

1. Executive Overview & Problem Formulation

Credit risk is the foundational risk assumed by lending institutions: the potential that a borrower will fail to meet their legal debt obligations. Modern banking demands systems that balance rigorous statistical discrimination with real-time operational monitoring, explainable machine learning, and secure artificial intelligence. This project establishes an institutional-grade credit monitoring platform built from 5 synthetic commercial bank tables (~10,000 customers, 14,685 loan accounts).

The solution delivers three core capabilities: (1) **High-integrity data ingestion & feature engineering** resolving relational fan-out traps and data quality defects; (2) **Supervised default prediction modeling** evaluating Logistic Regression and Random Forest architectures alongside Tree SHAP explainability and Population Stability Index (PSI) tracking; and (3) **Two embedded AI features** within an interactive Streamlit dashboard: a grounded natural-language assistant ('Ask the Portfolio') and a peer-comparative behavioral anomaly detection queue with human-in-the-loop audit logging.

2. Credit Risk & Machine Learning Demystified: Key Terms Explained Simply

To ensure complete transparency across technical and executive stakeholders, below is a plain-English explanation of core concepts and formulas used in this project:

- **Default & Probability of Default (PD):** A loan 'default' occurs when a borrower fails to pay installments as agreed for a sustained duration (conventionally 90+ days past due). The Probability of Default (PD) is a statistical number between 0% and 100% representing the model's estimate that an account will default within the observation window.
- **AUC (Area Under the ROC Curve):** AUC measures the model's ranking ability. *Intuitive Analogy:* If you randomly pick one borrower who defaulted and one borrower who paid on time, AUC is the probability that your model assigned a higher risk score to the defaulted borrower. An AUC of 0.50 is equivalent to flipping a coin (zero signal); 1.0 represents psychic perfection. In retail credit risk, an AUC of 0.65–0.75 is typical, robust, and represents significant commercial value.
- **KS Statistic (Kolmogorov-Smirnov):** The KS statistic measures the maximum percentage separation between the cumulative distribution of defaulters and non-defaulters across risk score deciles. *Intuitive Analogy:* Imagine lining up all defaulters from highest score to lowest, and doing the same for non-defaulters. KS is the widest gap between the two curves. A KS above 0.20 confirms the model cleanly separates good borrowers from distressed borrowers.
- **Gini Coefficient (2 * AUC - 1):** Directly derived from the Lorenz curve, Gini is the global banking standard metric under the Basel Capital Accords. It rescales AUC to a 0 to 1 scale (Gini = 2 * AUC - 1). A model with AUC 0.6609 has a Gini of 0.3217, confirming solid rank-ordering discrimination.
- **SHAP (SHapley Additive exPlanations):** Rooted in cooperative game theory (Nobel Prize in Economics), SHAP calculates the exact dollar or percentage contribution of each borrower attribute toward their final risk score. *Intuitive Analogy:* If an account has an 18% default risk while the portfolio average is 16%, SHAP tells you exactly which factors pushed it up (e.g. +4% from low income, +3% from missed payments) and which pushed it down (e.g. -5% from long employment tenure). This satisfies regulatory 'Adverse Action' notice requirements.
- **PSI (Population Stability Index):** PSI measures whether the distribution of credit scores has drifted over time compared to the training baseline. Values < 0.10 indicate a stable population (Green); 0.10 to 0.25 indicate moderate shift (Yellow, trigger review); >= 0.25 indicate severe structural drift (Red, halt automated approvals and re-train).
- **DTI (Debt-to-Income Ratio) & DPD (Days Past Due):** DTI is the ratio of total loan liabilities to borrower annual income (higher DTI = higher debt burden). DPD tracks how many days have elapsed since a scheduled payment was due without being paid in full (e.g., 30 DPD, 60 DPD, 90 DPD).

3. Data Ingestion Architecture & Data Quality Decisions

Commercial banking data is inherently messy and distributed across relational tables with differing grains. A critical failure mode in credit engineering is the **'Cartesian Fan-Out Trap'**: naively merging many-to-one tables (like monthly repayments or external credit lines) directly onto loan accounts. This corrupts row counts, multiplies exposures, and invalidates statistical calculations.

Table Grain Architecture:

Table Name	Source Grain	Target Grain	Aggregation & Join Strategy
loan_accounts.csv	1 row / loan (14,685)	Account Grain (Anchor)	Anchor fact table holding target_default (16.10% default rate).
customers.csv	1 row / customer (10,000)	Customer Grain	Left merge many_to_one on customer_id (100% match rate).
bureau_credit.csv	1 row / line (22,979)	Customer Grain	Group by customer_id -> merge many_to_one. 10% thin-files cleanly zero-filled.
repayment_history.csv	1 row / acct / mo (176k)	Account Grain	Group by account_id -> merge one_to_one across 12 monthly cycles.
collections_events.csv	1 row / event (8,635)	Account Grain	Group by account_id -> merge one_to_one for 4,278 delinquent facilities.

Explicit Data Quality Decisions & Remediation:

1. **Inconsistent Gender Labels:** ~6% of records in customers.csv were encoded as 'M' or 'F' instead of 'Male' and 'Female'. Resolved via standard mapping {'M': 'Male', 'F': 'Female'}. No rows dropped.
2. **Missing Profile Values:** years_employed, num_dependents, and years_at_residence each had 500 missing values (~5%). Added explicit binary missingness indicator flags (e.g. years_employed_missing_flag) to capture non-reporting risk signals, then imputed medians.
3. **Missing Interest Rates:** 294 loan accounts had null interest rates. Flagged with interest_rate_pct_missing_flag and median-imputed.
4. **Bureau Active Line Logic:** 11,243 bureau lines had null date_closed. This was correctly recognized as *active open facilities* (not missing data) and aggregated into bureau_active_lines.
5. **Bureau Write-Off Bug Fix:** Resolved a subtle regex bug where str.contains('write|settled') failed to match 'Written-Off'. Fixed with str.contains('written|write|settled'), restoring the critical prior write-off delinquency signal.
6. **Feature Catalog:** 59 engineered variables constructed across payment ratios (repayment_paid_ratio), delinquency trajectory (repayment_dpd_trend), bureau utilization (bureau_utilization), and leverage (dti_ratio, total_debt_to_income).

4. Portfolio Analytics & Top 5 Empirical Insights ('So What' Actions)

A risk analytics dashboard must not merely display charts; it must translate empirical distributions into concrete underwriting policies. Analysis of the 14,685 loans reveals five profound structural patterns:

1. Steep Income Inversion Gradient (4.0x Risk Multiplier) *[Annual Income]*

- **Empirical Finding:** Borrowers earning <$30k experience a 38.50% default rate (174 / 452 accounts), compared to just 9.53% for borrowers earning $100k+ (293 / 3,076 accounts).
- ■ **Strategic / Underwriting Impact ('So What'):** Policy Directive: Implement a hard Debt-to-Income cap of 35% for sub-$30k earners, mandate independent third-party income verification, and reduce uncollateralized lending limits for entry-level income cohorts.

2. Extreme Delinquency Among Unemployed Borrowers *[Employment Type]*

- **Empirical Finding:** Unemployed borrowers exhibit a 34.96% default rate (251 / 718 accounts), more than double the 14.97% observed in Salaried professionals (n=7,546).
- ■ **Strategic / Underwriting Impact ('So What'):** Policy Directive: Restrict unsecured personal loans and credit cards to applicants with active, verified employment. For existing unemployed accounts, trigger early forbearance and loan restructuring workflows before 30 DPD.

3. Severe Capital Concentration in High-Defaulting Mortgages *[Loan Product Type]*

- **Empirical Finding:** Mortgage facilities exhibit the highest product default rate at 24.65% (353 / 1,432 accounts), while carrying an average balance of $219,605 ($314.5M original balance). Auto Loans default at 13.88%.
- ■ **Strategic / Underwriting Impact ('So What'):** Policy Directive: Because mortgage loss severity is driven by high ticket size, conduct immediate Loan-to-Value (LTV) appraisal audits, tighten property collateral haircut guidelines, and review broker origination channels.

4. Young Borrower Vulnerability (18–25 Age Cohort) *[Age Cohort]*

- **Empirical Finding:** Borrowers aged 18–25 show a 22.16% default rate (427 / 1,927), over double the 10.56% observed in mature borrowers aged 56+ (n=833).
- ■ **Strategic / Underwriting Impact ('So What'):** Policy Directive: Younger borrowers lack credit seasoning and savings buffers. Implement 'starter' credit lines with automated step-up limit reviews contingent on 6 consecutive on-time monthly payments.

5. Compounding Cross-Cut: Unemployed x Low Income Spike (44.1% Default Rate) *[Employment x Income Cross-Cut]*

- **Empirical Finding:** In the cross-dimensional intersection of Unemployed borrowers earning <$30k, default rate reaches 44.10% (229 accounts). Conversely, Salaried borrowers earning $100k+ default at only 8.99% — an 8.9x risk divergence across quadrants.
- ■ **Strategic / Underwriting Impact ('So What'):** Policy Directive: Implement bivariate underwriting cut-offs. Automatically knock out loan applications combining <$30k income with Unemployed or Contract status unless secured by 100% liquid cash collateral.

5. Supervised Default Prediction Modeling & Discrimination Results

To establish a reliable estimate of default probability (PD), we evaluated a regularized Logistic Regression baseline against Random Forest and LightGBM machine learning architectures. The model was trained on a **70/30 stratified holdout split** (10,279 train / 4,406 test accounts) using all 59 engineered features.

Model Architecture	Train AUC	Test AUC	Test KS	Test Gini	Overfit Gap	Deployment Status
Random Forest Classifier	0.9150	**0.6609**	**0.2342**	**0.3217**	0.2541	**Primary Production Model**
Logistic Regression Baseline	0.7009	**0.6565**	**0.2308**	**0.3131**	0.0444	Regulatory Baseline
LightGBM Classifier	0.9074	0.6453	0.2114	0.2906	0.2621	Challenger Benchmark

Top 5 Predictive Drivers (Random Forest Gini Importance & Tree SHAP):
1. annual_income (0.0644): Affordability anchor; lower income severely diminishes repayment cushion.
2. repayment_mean_dpd (0.0473): Primary behavioral delinquency signal; past delinquency strongly predicts future charge-off.
3. years_employed (0.0449): Proxy for job tenure and continuous cash flow stability.
4. total_debt_to_income (0.0426): Aggregate multi-institutional indebtedness relative to capacity.
5. bureau_mean_dpd (0.0398): External credit bureau track record across third-party lenders.

6. Dual Embedded AI Features & Enterprise Governance Blueprint

A core innovation of this platform is the integration of two distinct AI systems directly embedded into the decision workflows:

AI Feature 1: 'Ask the Portfolio' (Grounded LLM Assistant)

• **Mechanism:** Risk analysts submit natural language questions in Tab 4. The application compiles an on-the-fly JSON evidence payload from the currently filtered cohort (containing slice account counts, defaults, segment default rates, and balances). This JSON is transmitted via HTTPS to Groq Cloud (openai/gpt-oss-20b) or OpenAI (gpt-4o-mini).
• **Zero-PII Guarantee:** Under no circumstances are individual customer IDs, names, or row-level records sent to external APIs. Only aggregate counts and percentages are transmitted.
• **Anti-Hallucination Guardrails:** The system prompt explicitly commands the model to use *only* the provided numbers, state denominators, cite source tables, and explicitly decline out-of-scope questions (macro forecasts, competitors).
• **Deterministic Fallback:** If no API key is provided, the platform transparently falls back to an internal analytical query engine that calculates exact figures locally.

AI Feature 2: Behavioral Anomaly Detection & Review Queue

• **Mechanism:** Combines Scikit-Learn IsolationForest (contamination=0.03) with peer-group comparative benchmarks (Loan Type x Income Tier medians) and trajectory heuristics (DPD surges).
• **Transparent Reason Attribution:** Rather than an opaque black-box score, every flagged account is assigned human-readable triggers (e.g. 'Peak DPD 60d vs peer median 8d', '1 prior charged-off bureau line', 'DTI ratio 8.0x').
• **Human-in-the-Loop Audit Trail:** Risk analysts review flagged accounts in Tab 5, investigate profile context, and submit formal **Confirm / Dismiss / Escalate** decisions with case notes. Decisions persist durably to data/processed/analyst_review_decisions.json.

Mandatory AI Governance Specification Matrix:

AI Feature	Trigger	False Positive Risk	False Negative Risk	Human-in-the-Loop Step
Ask the Portfolio (LLM)	Analyst enters question and clicks 'Submit Query'.	LLM misinterprets slice denominator or asserts speculative causation.	LLM excessively declines a valid question inferable from slice data.	Analyst validates figures against dashboard KPI cards before policy memos.
Anomaly Detection	Automated background calculation on filtered cohort.	Prime borrower with irregular seasonal cash flow flagged as distressed.	Gradual delinquency deteriorating just beneath contamination limit.	Analyst investigates case dossier and logs persistent Confirm / Dismiss decision.

7. Interactive Dashboard Design & User Workflows

The Streamlit dashboard (app/app.py) is organized into 6 specialized operational tabs:

- **Tab 1: Portfolio Overview:** Executive summary cards (accounts, default rate %, exposure, origination volume), product default rate bars, risk band pie chart (<10%, 10–25%, ≥25%), and historical monthly vintage curves.
- **Tab 2: Segmentation & Insights:** Socio-demographic profiling across Income, Employment, Region, Age, Loan Type; plus the interactive **Employment Type x Income Band** cross-tab heatmap and the 5 empirical insight cards.
- **Tab 3: Account Risk & SHAP:** Searchable borrower dossier displaying predicted PD, loan terms, bureau history, 12-month repayment performance, and a **local risk driver waterfall chart** explaining individual risk contributions.
- **Tab 4: Ask the Portfolio (LLM):** Grounded conversational assistant with query preset pills, live API connectivity status, exact numerical citations, and out-of-scope refusal logic.
- **Tab 5: Anomaly Detection Queue:** Review queue of flagged behavioral outliers, highlighted reason badges, peer median comparisons, interactive decision recording, and a live decision audit table.
- **Tab 6: Model Monitoring & Governance:** Holdout discrimination metrics table, top 15 global feature importance chart, decile calibration (Expected vs Actual defaults), and PSI stability monitoring specs.

8. Post-Deployment Monitoring & Enterprise Production Roadmap

In commercial banking, model deployment is merely Day 1. Regulatory compliance under **Federal Reserve SR 11-7 / OCC 2011-12 (Supervisory Guidance on Model Risk Management)** requires continuous monitoring for distribution drift and performance degradation.

1. **Population Stability Index (PSI) Protocol:** Monthly automated score distribution comparison against training baseline. Current simulated test PSI is 0.0250 (Green: < 0.10). If PSI crosses 0.10, warning alerts trigger; if >= 0.25, automated approvals are paused.
2. **Discrimination Degradation Floors:** Automated monitoring tracks rolling 3-month holdout performance. If Test AUC drops below 0.65 or KS drops below 0.20, model re-estimation is automatically initiated.
3. **Decile Calibration Monitoring:** Expected-to-Actual (E/A) default ratios tracked by decile; deviations beyond ±20% trigger credit reserve adjustments.

4. Bank IT & Data Engineering Production Requirements:

- *Point-in-Time Feature Store* (e.g. Feast / Hopsworks): To prevent future data leakage and guarantee point-in-time feature consistency across training and live scoring.
- *Enterprise Secrets Vault* (e.g. AWS Secrets Manager / Azure Key Vault / HashiCorp Vault): Secure storage and automated rotation of LLM API credentials without file-based .env exposure.
- *Durable Audit Database* (PostgreSQL / Snowflake): Scalable ACID-compliant persistence for analyst review decisions, timestamps, and justification notes for regulatory auditability.
- *Model Risk Governance (MRM) Sign-Off:* Independent validation review, developmental evidence documentation, and approval under supervisory SR 11-7 standards.